Incompressible Euler Flow: Turbulence vs Blow-up

Johan Hoffman
KTH Royal Institute of Technology, Stockholm
Claes Johnson
KTH Royal Institute of Technology, Stockholm

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Abstract

All physical Euler flow is a form of bluff body flow. Without an obstruction in the flow, and with slip on the boundary, nothing happens: a uniform stream in a straight pipe solves the equations for ever, there being no shear to be unstable and no viscosity to make a wall layer. Euler with slip has no mechanism for generating structure, so everything that makes a flow a flow enters because the fluid must go around something. We therefore take flow past a body as the only setting in which the equations describe a fluid. There potential flow is a solution and it is unstable, but not everywhere. Attachment, on the forward surface where the flow is accelerated, is stable. Separation, on the rear surface where it is retarded, is not: the opposing flow there is unstable to spanwise perturbation and generates streamwise vorticity, which stretches, breaks down and feeds a cascade. The flow that results has bounded velocity and unbounded velocity gradients, is Hölder continuous with exponent about 1/3, and dissipates energy at a rate that does not vanish — a dissipative weak solution in the sense of Onsager. This is what flow past a body does, and the distinction between streamlined and bluff bodies does not survive it: every body separates, so all flow past a body is bluff body flow, and there is no potential flow regime at any Reynolds number.

We then compare this with the programme that constructs finite-time singularities of Euler, which has succeeded in several settings. Our conclusion is that *all physical Euler flow is turbulent and the blow-up solutions are not physical.* They are theorems and we do not dispute them; they are not flows. The reason is a matter of function space. The singular solutions are constructed in $C^{1,\alpha}$, far above Onsager's exponent 1/3, where energy is necessarily conserved; a solution that conserves energy has no cascade available to it, and nowhere to put a growing gradient but concentration. Physical flow lies below the exponent, where dissipation is available, and takes it. Blow-up is what Euler does when it cannot dissipate; turbulence is what it does when it can, and a fluid always can. The constructions are moreover posed either on the whole space, where there is no body and so no separation, or with a symmetry and a boundary that supply the concentration by hand — and stability within a symmetry class is not genericity, least of all when the physical instability is the one that breaks the symmetry.

1 All physical Euler flow is bluff body flow

We begin with the claim the title rests on, because it is not a summary of what follows but the reason there is anything to follow.

All physical Euler flow is a form of bluff body flow. Without an obstruction in the flow, and with slip on the boundary, nothing happens at all. Take a straight pipe and a uniform

stream along it. With $\nu = 0$ and slip on the wall, u constant and p constant solve the Euler equations exactly, and go on solving them for ever. There is no shear, so there is nothing to be unstable; there is no wall layer, because there is no viscosity to make one; and there is nothing for the fluid to flow past. The solution is the initial data, unchanged, for all time. One may compute it, but there is nothing in it to compute.

This is not a curiosity about pipes. It is the general situation. The Euler equations with slip have no internal mechanism for generating structure: vorticity is transported and stretched but not created, so a flow that begins irrotational and meets nothing stays irrotational, and a flow that meets nothing at all stays whatever it was. Everything that makes a flow a flow — vorticity, separation, a wake, drag, turbulence — enters because the fluid is obliged to go around something. The obstruction is not one case among many. It is the only source.

And once there is an obstruction, there is a forward surface where the flow is accelerated and a rear surface where it is retarded, because the pressure must return to what it was. The first is stable and the second is not, as Section 2 sets out, and from the instability of the second everything else follows. That is bluff body flow. A streamlined body retards the flow more gently over a longer rear surface and so separates later and leaves a narrower wake, but it separates, and the mechanism is the same one. There is no third kind of body and no flow without a body worth the name. Hence the claim, and hence also the shape of this paper: the setting is not a special case chosen for convenience, it is the only setting in which the Euler equations describe anything.

1.1 Euler with slip, past a body

The incompressible Euler equations

$$\dot{u} + (u \cdot \nabla)u + \nabla p = 0, \qquad \nabla \cdot u = 0, \tag{1}$$

are posed in a domain Ω exterior to a body B, with $u \to U$ at infinity and the *slip* condition

$$u \cdot n = 0 \quad \text{on} \quad \partial B, \tag{2}$$

no condition being imposed on the tangential velocity.

Neither of these is a free choice, and both are worth a word, because the first is routinely treated as one and the second is usually not noticed at all.

Why slip. Because the viscosity is zero. The slip condition (2) is not a modelling choice to be weighed against no-slip; it is the only boundary condition the Euler equations admit on an impermeable wall. The wall must not be crossed, which fixes $u \cdot n$; nothing in (1) determines the tangential velocity there, and imposing $u = 0$ would over-determine the problem. No-slip belongs to Navier–Stokes, where the viscous term is second order and carries the extra condition, and the boundary layer it produces has thickness $\sim Re^{-1/2}$ and disappears in the inviscid limit. At $\nu = 0$ there is no boundary layer to be thin. There is slip.

This should be said because the drag of a body is habitually attributed to the boundary layer, and in the present setting there is none. Whatever drag appears below is produced without viscosity and without a wall layer, by the instability of Section 2 alone. That the resulting computations give the right numbers — mean drag within a few per cent for the

circular cylinder, the separation line in the observed place, the lift curve and the stall for wings [4, 3] — is then evidence that nothing essential was left out. The turbulent Euler solutions obtained this way on coarse meshes, including the drag crisis, are reported in [6].

Why past a body. The Cauchy problem for (1) on $\mathbb{R}^3$ or on the torus is a perfectly well-posed mathematical question and we do not suggest otherwise. We observe only that it is not a question about a fluid. A fluid flows past something; with no boundary there is no attachment and no separation, hence none of the mechanism described below, and what is being studied is an initial-value problem for a quasilinear system rather than a flow. This matters later, because several of the singularity constructions are posed exactly there.

2 d'Alembert: attachment is stable, separation is not

Potential flow — the irrotational solution of (1) with (2) — exists for any body, is unique, and exerts zero drag. That is d'Alembert's paradox, and for a century and a half the standard resolution located the missing drag in viscosity and the boundary layer. The resolution we gave in [2] locates it elsewhere: potential flow is a solution of the equations, and it is *unstable*, so it is not observed. Nothing about viscosity is required to say so.

What matters for everything that follows is that the instability is not uniform over the body. Divide the surface by the sign of the streamwise pressure gradient.

On the forward surface the flow is *accelerated*: the pressure gradient is favourable, fluid elements are stretched along the flow direction and compressed across it, and a perturbation to the velocity is correspondingly stretched and weakened. Attachment is stable. This is why the forward stagnation point and the flow over the front of a body are seen in experiment exactly where potential theory puts them, and why potential flow is an excellent description of the forward surface.

On the rear surface the flow is *retarded*: the pressure gradient opposes the motion, and potential flow asks opposing streams to meet at a rear stagnation point and separate smoothly. That configuration is unstable. Perturbing it in the spanwise direction, the opposing flows do not meet on a line but interleave, and the result is a row of counter-rotating vortex tubes aligned with the flow. The growth is exponential with rate set by $|\nabla u|$ at separation, so it is fast on the convective time scale, and it is not a slow accumulation that a sufficiently careful experiment could avoid.

The asymmetry is the whole of the resolution. The drag is zero in potential flow because the pressure recovers on the rear exactly as it fell on the front. The instability destroys the recovery and only the recovery: the front is as potential theory says, the rear is not, and the difference is the drag.

3 What the instability generates

The perturbation that grows at separation is specific, and its form is what connects a local instability to a global cascade.

The unstable mode generates *streamwise* vorticity: vortex tubes aligned with the flow, in

counter-rotating pairs, originating at the separation line. This is not incidental. A vortex tube aligned with the flow sits in the strain field of the flow that produced it, and that field stretches it. Stretching a vortex tube intensifies its vorticity and thins its core, which is to say it moves energy from the scale of the body to a smaller scale, and it does so without any help. The stretched tube is then itself unstable, breaks down into finer tubes, and the process repeats.

What this gives is a cascade with an origin. In the usual presentation the Kolmogorov cascade is a statistical hypothesis about an inertial range, with the large scales stirred by some unspecified mechanism. Here the mechanism is named and located: it is the separation instability, it acts at the rear of the body, and the first generation of the cascade is the counter-rotating streamwise tubes it produces. The inertial range is what happens to them.

Energy enters this process at a rate set by the body and the free stream, $\varepsilon \sim U^3/L$, and nothing in the chain depends on viscosity. Each generation passes its energy down, and the rate is the same at every scale because there is nowhere else for it to go.

4 No streamlined exception

Section 1 claimed that all physical Euler flow is bluff body flow, and the instability of Section 2 is what makes the claim bite. The objection to meet is the conventional division of bodies into bluff and streamlined, the first separating and leaving a wide wake, the second supposedly not. The division does not survive the account above.

Every body separates, because every body is finite and the flow that accelerates over its forward surface must be retarded over its rear. A streamlined body is one whose rear surface retards the flow gently and whose separation is therefore delayed and its wake narrow, and this is a difference of degree in the size of the unstable region, not a difference in kind. There is no shape for which the rear pressure gradient is favourable everywhere, because the pressure must return to its free-stream value.

This is a record, not an assertion. The statements above are not predictions about what would happen to a perturbed potential solution. Taking $\nu \to 0$ in the General Galerkin method of [1] gives a solver for (1) whose dissipation does not vanish with the viscosity, and what it produces, started from smooth data and with no model imposed to keep it smooth, is not potential flow. A bluff body in this limit shows potential flow modified by a three-dimensional rotational separation, and that separation carries the drag. The turbulent Euler solutions so obtained, including the drag crisis, are reported in [6]; the same computation applied to a wing gives lift and drag with no circulation condition imposed by hand [3].

Seen this way d'Alembert's paradox is a mistaken limit rather than a missing force. It arises from taking the smooth potential solution to be the vanishing-viscosity limit of a real flow. The correct limit is turbulent, and it has drag.

Two consequences follow and both are testable. There is no potential flow regime: potential flow is unstable past every body at every Reynolds number, so the limit $Re \to \infty$ does not recover it, and the inviscid limit of a real flow is turbulent and not smooth. And the same mechanism accounts for a wing: a wing is a slender bluff body whose separation is organised rather than suppressed, and the lift it generates comes about because the instability acts at the trailing edge and on the upper surface in a way the geometry controls. That is the account

of flight given in [3], and it is the same instability as the cylinder's.

5 What the physical solution is

The flow produced by this process is not a classical solution of (1), and it is worth being exact about what it is instead, since the comparison in Section 6 turns on it.

Energy is bounded: the velocity is bounded by the free stream and the domain is fixed, so $\|u(t)\|_{L^2}$ does not grow. Velocity *gradients* are not bounded. The cascade refines without limit, since there is no viscosity to stop it, and the velocity difference across a separation ℓ behaves as

$$\delta u(\ell) \sim (\varepsilon \ell)^{1/3}, \qquad \text{so} \qquad \frac{\delta u(\ell)}{\ell} \sim \varepsilon^{1/3} \ell^{-2/3} \to \infty. \tag{3}$$

The first relation is Kolmogorov's [8], and read as a statement about the velocity field it says that u is Hölder continuous with exponent $1/3$ and no better.

That exponent is not a detail. Onsager observed [10] that a weak solution of Euler conserves energy if its velocity is Hölder continuous with exponent greater than $1/3$, and that below the exponent it need not. The first half was proved by Constantin, E and Titi [11]; the second half — that dissipative weak solutions below the exponent exist — was established through the convex-integration programme of De Lellis and Székelyhidi and collaborators [13, 14] and completed by Isett [15]. Eyink's analysis [12] identifies the mechanism as a local energy flux that survives the inviscid limit.

Onsager's threshold is now settled in both directions — above the exponent energy is conserved, below it dissipating solutions exist and have been constructed — so the class we are describing, bounded and not smooth and dissipating, is not a conjecture about what solutions might do. It is a category with occupants.

So the physical solution is a *dissipative weak solution*: bounded velocity, unbounded gradient, Hölder exponent at the Onsager value, and energy dissipated at a rate ε that does not vanish although the equation contains no dissipative term. The dissipation is anomalous in the technical sense and entirely ordinary in the physical one — it is the drag, and it has been measured since Taylor [9].

That the dissipation in external flow past a body is inertial rather than viscous, and survives the inviscid limit, is not only our claim: it has been pursued independently within the Onsager framework for flow around a smooth body [7], where the momentum dissipation is located at the boundary and in the wake. We take the agreement as support for the mechanism and note the difference in route — that work argues from the viscous side and takes a limit, and the account here starts at $\nu = 0$ and never has a boundary layer to dispose of.

It should be said plainly that we do not derive (3) from (1). What we claim is the mechanism that puts a flow into that class and keeps it there: separation is unstable, the instability generates streamwise vorticity, stretching carries it down the scales, and the cascade terminates nowhere because nothing terminates it.

6 The constructed singular solutions

Against this stands a programme, now substantially successful, that constructs solutions of (1) whose velocity becomes infinite in finite time. Luo and Hou exhibited a numerical scenario with boundary [16]; Elgindi proved finite-time singularity formation for axisymmetric flow without swirl at $C^{1,\alpha}$ regularity [17]; Chen and Hou gave a computer-assisted proof for the 2D Boussinesq and 3D axisymmetric Euler equations with boundary and smooth initial data [18]; and the regularity has since been pushed to $C^{1,1/3}$ [19], with stability on the whole space also claimed [20].

We have been here before, and it is worth saying where we stood. In [5] we studied blowup of incompressible Euler solutions computationally and concluded that what the equations produce is not a point singularity but a breakdown of the flow into turbulence — the velocity staying bounded while the gradients do not. The present paper is that conclusion stated in the language the subject has since acquired, with Onsager's exponent as the thing that separates the two outcomes, and with the separation instability named as the mechanism that drives a flow across it.

These are theorems and we do not dispute them. We claim that they are not flows. All physical Euler flow is turbulent, by the mechanism of Sections 2–3, which acts on every body and has no exceptions; the singular solutions are solutions of the equations that no fluid realises. The reason is not rhetorical but a matter of which function space each account occupies, and it can be stated exactly.

The regularity is the point. Every one of these constructions lives in $C^{1,\alpha}$: the velocity is continuously differentiable with Hölder continuous derivatives. That is far *above* Onsager's exponent. A $C^{1,\alpha}$ velocity is Lipschitz, hence Hölder continuous with every exponent up to 1, hence well above $1/3$ — and by [11] such a solution *conserves energy exactly*, on the whole interval up to the singular time.

A solution that conserves energy has no cascade available to it. The energy present cannot be passed down the scales and removed; it must remain, and the only way to accommodate a growing gradient while conserving the integral is to concentrate. The singularity is what that concentration becomes — it is forced, and it is forced by the smoothness. That is the sense in which these solutions are not physical. A fluid is not confined to $C^{1,\alpha}$. It is free to roughen, the separation instability makes it do so immediately, and once it is below the Onsager exponent the cascade is open and the energy leaves. The concentration never has to happen, so it does not.

> Blow-up is what Euler does when it cannot dissipate. Turbulence is what it does when it can.

What regularity theory can reach. It is worth adding why the question is not settled by bounding things. The analytic tradition produces conditional results of the form: if the velocity is bounded by A, then its derivatives are bounded by an expression exponential in A iterated several times. As theorems these are real. As statements about a fluid they are empty — a bound of size $\exp\exp\exp A$ is not one any flow can be said to respect, and no measurement could distinguish it from no bound at all. A Hölder-1/3 field is neither differentiable nor

unbounded, and that is exactly the object regularity theory has no way to reach: it is not looking for a category between smooth and singular, because the dichotomy it works in has no name for one.

And the setting is the point too. The constructions are posed either on the whole space, where by Section 1 there is no body, no slip condition and no separation — so no part of the mechanism of Sections 2–3 is present — or in a cylinder with a boundary and an imposed symmetry, where the symmetry is what forces the flow onto the axis and supplies the concentration. In neither case is a body being passed. The data are special in the strong sense that they are chosen so that the self-similar profile exists, and the proof is the verification that it does.

On stability. The strongest form of the objection to the above is that blow-up has been shown stable, and stability is what distinguishes the generic from the constructed. We would observe that stability is always stability *within* a class, and that the classes here are symmetry classes: axisymmetric, or axisymmetric with swirl, or with a prescribed sign structure. A perturbation that breaks the symmetry is not in the class, and it is exactly the symmetry-breaking spanwise perturbation that in Section 2 is the unstable one. Stability within the symmetric class and instability to spanwise perturbation are not in conflict; they are the same statement seen twice, and it is the second that decides what a flow does.

7 Summary

Euler flow past a body with slip has an unstable separation and a stable attachment. The separation instability generates streamwise vorticity, the vorticity is stretched and broken down, and the result is a cascade which has an origin and no termination. The flow so produced has bounded velocity, unbounded gradients, Hölder exponent 1/3, and dissipates at a rate independent of any viscosity: a dissipative weak solution in Onsager's sense. Every body separates, so the streamlined case is the bluff case with a smaller unstable region, and there is no potential flow regime at any Reynolds number.

The singular solutions constructed in recent years are solutions of the same equations in a different function space, one smooth enough that energy is conserved and therefore smooth enough that a cascade is unavailable. They are posed where there is no body to separate from, and their stability is stability within a symmetry the physical instability breaks. They describe what the Euler equations can be made to do under a smoothness no fluid maintains. The account above describes what fluids do. All physical Euler flow is turbulent; the blow-up solutions are not physical.

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